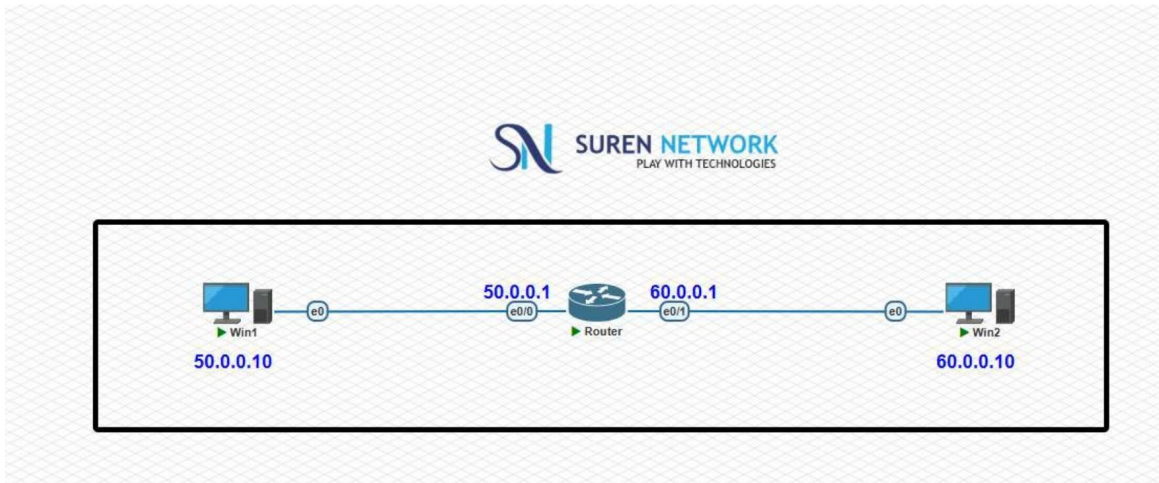


CCNA Practical Lab Report

Single Router — Static Routes Configured on End Hosts

Two hosts (Win1 — 50.0.0.10, Win2 — 60.0.0.10) sit on opposite sides of a router with no default gateway set on either PC. Connectivity was achieved by adding a manual static route on each host pointing to the router as next hop.

Topology



Router Configuration

Ethernet0/0 — 50.0.0.1/8

```
Router#
Router#enable
Router#config term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int ethernet0/0
Router(config-if)#ip add 50.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
*Aug 22 14:27:49.120: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip int br
Interface                IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0              50.0.0.1       YES manual up          up
Ethernet0/1              unassigned     YES TFTP  up          up
Ethernet0/2              unassigned     YES TFTP  down        down
Ethernet0/3              unassigned     YES TFTP  down        down
Router#
```

Ethernet0/1 — 60.0.0.1/8

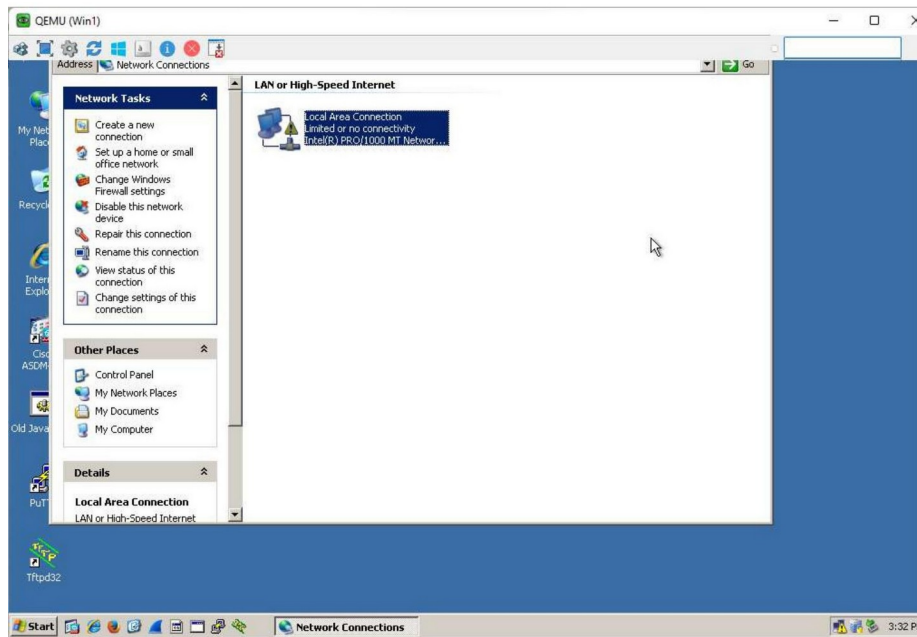
```

Router
Router#enable
Router#config term
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int ethernet0/1
Router(config-if)#ip add 60.0.0.1 255.0.0.0
Router(config-if)#
*Aug 22 14:29:57.673: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
Router(config-if)#ip add 60.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
*Aug 22 14:30:40.039: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip int br
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              50.0.0.1        YES manual up            up
Ethernet0/1              60.0.0.1        YES manual up            up
Ethernet0/2              unassigned      YES TFTP  down          down
Ethernet0/3              unassigned      YES TFTP  down          down
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#

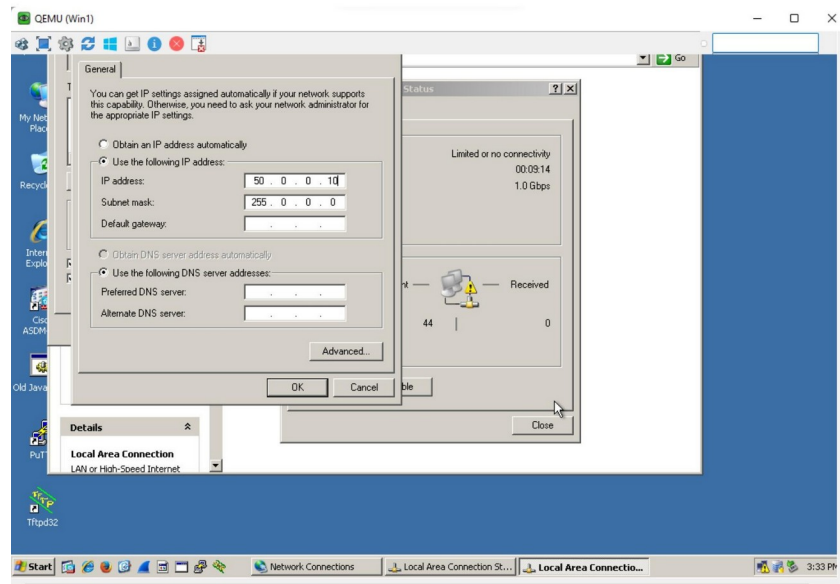
```

PC IP Configuration

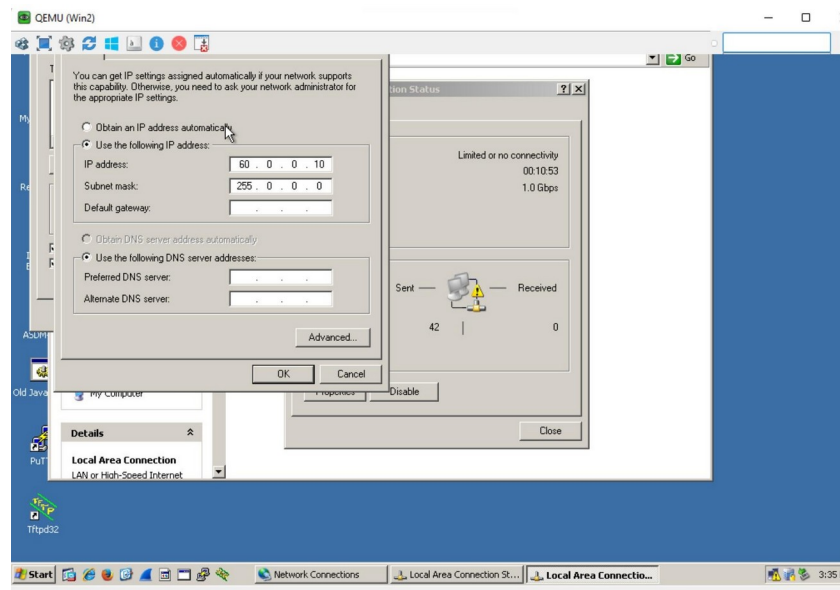
Win1 — Local Area Connection (no gateway set)



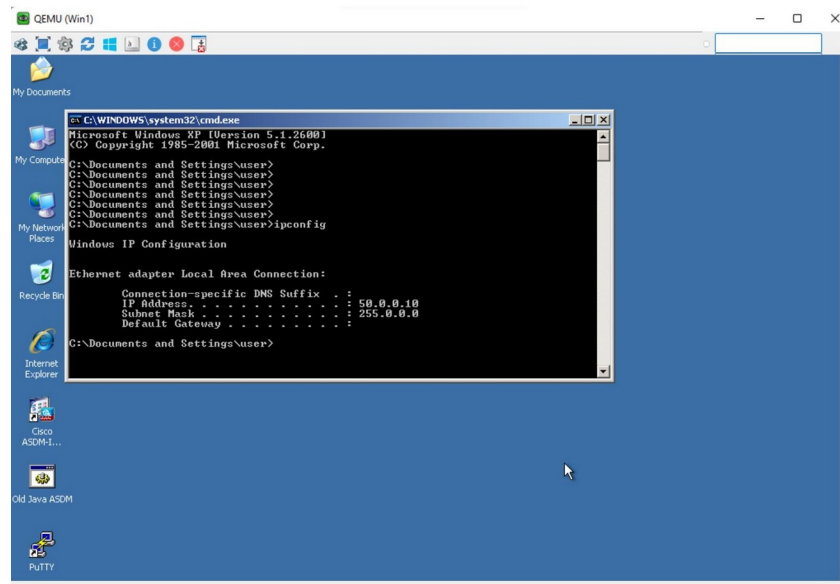
Win1 — 50.0.0.10 / 255.0.0.0



Win2 — 60.0.0.10 / 255.0.0.0

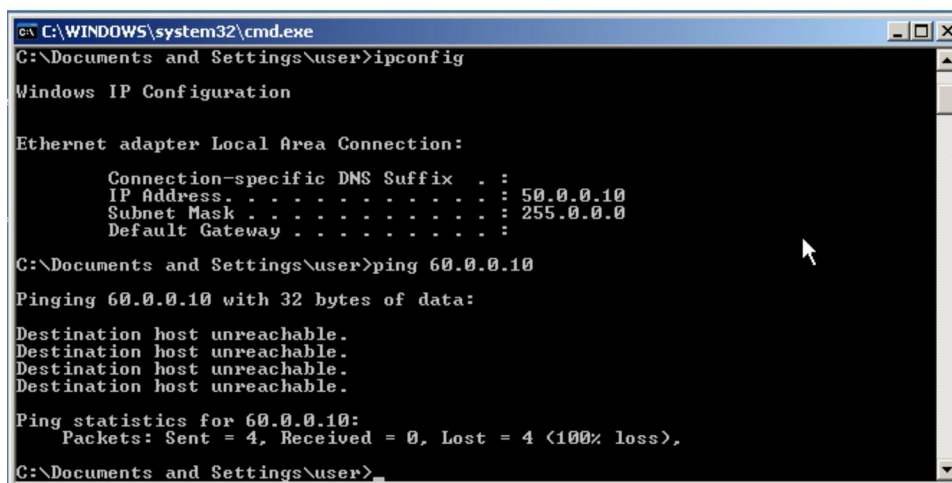


Win1 ipconfig confirms 50.0.0.10, no default gateway

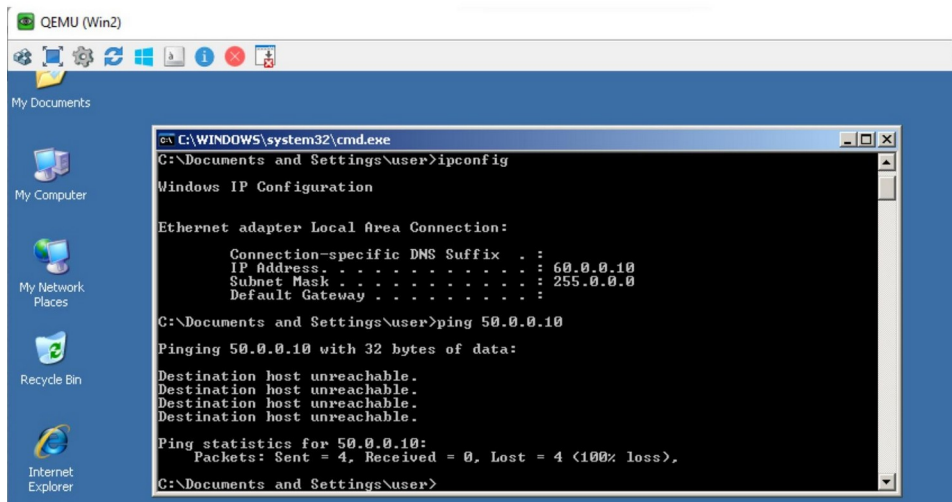


Initial Test — No Route, No Connectivity

Win1 → 60.0.0.10: Destination host unreachable

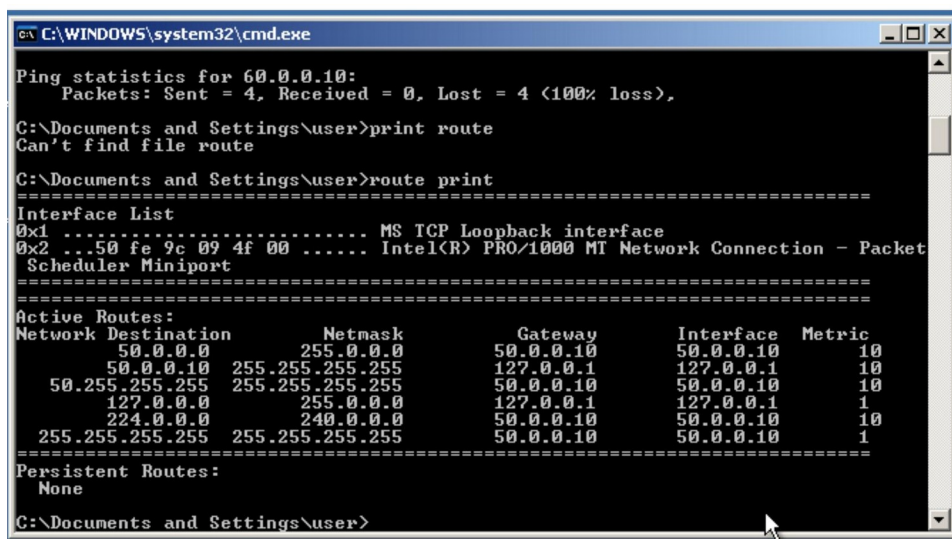


Win2 → 50.0.0.10: Destination host unreachable



Adding a Static Route on Win1

route print (before) — no route to 60.0.0.0



```

C:\WINDOWS\system32\cmd.exe

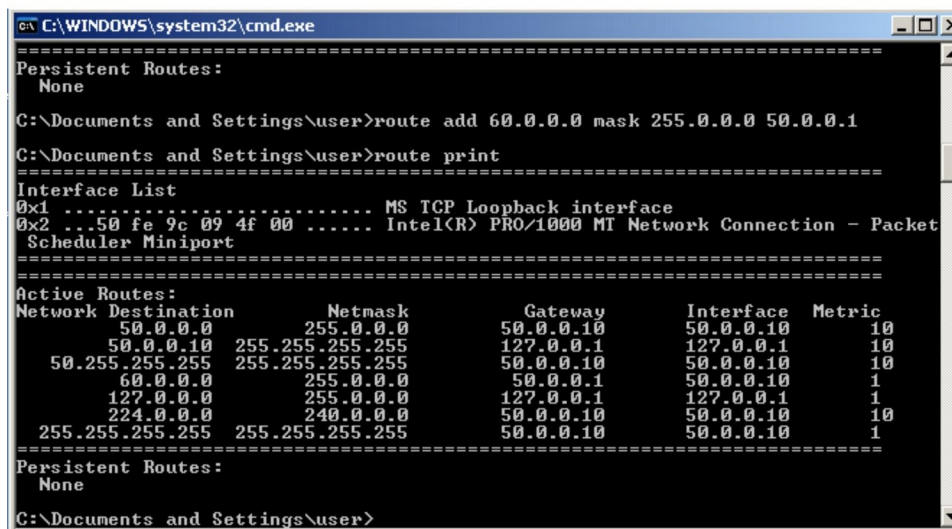
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>print route
Can't find file route

C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
-----
50.0.0.0                   255.0.0.0         50.0.0.10         50.0.0.10         10
50.0.0.10                  255.255.255.255   127.0.0.1         127.0.0.1         10
50.255.255.255             255.255.255.255   50.0.0.10         50.0.0.10         10
127.0.0.0                  255.0.0.0         127.0.0.1         127.0.0.1         1
224.0.0.0                  240.0.0.0         50.0.0.10         50.0.0.10         10
255.255.255.255           255.255.255.255   50.0.0.10         50.0.0.10         1
=====
Persistent Routes:
    None

C:\Documents and Settings\user>
  
```

route add 60.0.0.0 mask 255.0.0.0 50.0.0.1 → route print (after)



```

C:\WINDOWS\system32\cmd.exe

=====
Persistent Routes:
    None

C:\Documents and Settings\user>route add 60.0.0.0 mask 255.0.0.0 50.0.0.1

C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
-----
50.0.0.0                   255.0.0.0         50.0.0.10         50.0.0.10         10
50.0.0.10                  255.255.255.255   127.0.0.1         127.0.0.1         10
50.255.255.255             255.255.255.255   50.0.0.10         50.0.0.10         10
60.0.0.0                   255.0.0.0         50.0.0.1          50.0.0.10         1
127.0.0.0                  255.0.0.0         127.0.0.1         127.0.0.1         1
224.0.0.0                  240.0.0.0         50.0.0.10         50.0.0.10         10
255.255.255.255           255.255.255.255   50.0.0.10         50.0.0.10         1
=====
Persistent Routes:
    None

C:\Documents and Settings\user>
  
```

Ping still times out (Win2 has no return route yet)

```

C:\WINDOWS\system32\cmd.exe
=====
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
50.0.0.0                255.0.0.0        50.0.0.10       50.0.0.10        10
50.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1        10
50.255.255.255         255.255.255.255  50.0.0.10       50.0.0.10        10
60.0.0.0                255.0.0.0        50.0.0.1        50.0.0.10        1
127.0.0.0              255.0.0.0        127.0.0.1       127.0.0.1        1
224.0.0.0              240.0.0.0        50.0.0.10       50.0.0.10        10
255.255.255.255        255.255.255.255  50.0.0.10       50.0.0.10        1
=====
Persistent Routes:
None

C:\Documents and Settings\user>ping 60.0.0.10

Pinging 60.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>

```

Adding a Static Route on Win2

route add 50.0.0.0 mask 255.0.0.0 60.0.0.1

```

C:\WINDOWS\system32\cmd.exe
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 86 5f 09 50 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet
Scheduler Miniport
=====
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
60.0.0.0                255.0.0.0        60.0.0.10       60.0.0.10        10
60.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1        10
60.255.255.255         255.255.255.255  60.0.0.10       60.0.0.10        10
127.0.0.0              255.0.0.0        127.0.0.1       127.0.0.1        1
224.0.0.0              240.0.0.0        60.0.0.10       60.0.0.10        10
255.255.255.255        255.255.255.255  60.0.0.10       60.0.0.10        1
=====
Persistent Routes:
None

C:\Documents and Settings\user>route add 50.0.0.0 mask 255.0.0.0 60.0.0.1
C:\Documents and Settings\user>

```

route print confirms 50.0.0.0 via 60.0.0.1

```

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>route add 50.0.0.0 mask 255.0.0.0 60.0.0.1
C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 86 5f 09 50 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet
Scheduler Miniport
=====
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
50.0.0.0                255.0.0.0        60.0.0.1        60.0.0.10        1
60.0.0.0                255.0.0.0        60.0.0.10       60.0.0.10        10
60.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1        10
60.255.255.255         255.255.255.255  60.0.0.10       60.0.0.10        10
127.0.0.0              255.0.0.0        127.0.0.1       127.0.0.1        1
224.0.0.0              240.0.0.0        60.0.0.10       60.0.0.10        10
255.255.255.255        255.255.255.255  60.0.0.10       60.0.0.10        1
=====
Persistent Routes:
None

C:\Documents and Settings\user>D

```

Connectivity Verified

Win1 routing table with 60.0.0.0 route in place

```

C:\WINDOWS\system32\cmd.exe
=====
Persistent Routes:
None

C:\Documents and Settings\user>route add 60.0.0.0 mask 255.0.0.0 50.0.0.1

C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 fe 7c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet
Scheduler Miniport
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
50.0.0.0                   255.0.0.0        50.0.0.10       50.0.0.10       10
50.0.0.10                  255.255.255.255  127.0.0.1       127.0.0.1       10
50.255.255.255             255.255.255.255  50.0.0.10       50.0.0.10       10
60.0.0.0                   255.0.0.0        50.0.0.1        50.0.0.10       1
127.0.0.0                  255.0.0.0        127.0.0.1       127.0.0.1       1
224.0.0.0                  240.0.0.0        50.0.0.10       50.0.0.10       10
255.255.255.255           255.255.255.255  50.0.0.10       50.0.0.10       1
=====
Persistent Routes:

```

Win1 → 60.0.0.1 (router): ping successful

```

C:\WINDOWS\system32\cmd.exe
=====
50.0.0.10 255.255.255.255 127.0.0.1
50.255.255.255 255.255.255.255 50.0.0.10
60.0.0.0 255.0.0.0 50.0.0.1
127.0.0.0 255.0.0.0 127.0.0.1
224.0.0.0 240.0.0.0 50.0.0.10
255.255.255.255 255.255.255.255 50.0.0.10
=====
Persistent Routes:
None

C:\Documents and Settings\user>ping 60.0.0.1

```

Full ping statistics: 0% loss

```

C:\WINDOWS\system32\cmd.exe
=====
50.0.0.10 255.255.255.255 127.0.0.1 127.0.0.1 10
50.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 10
60.0.0.0 255.0.0.0 50.0.0.1 50.0.0.10 1
127.0.0.0 255.0.0.0 127.0.0.1 127.0.0.1 1
224.0.0.0 240.0.0.0 50.0.0.10 50.0.0.10 10
255.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 1
=====
Persistent Routes:
None

C:\Documents and Settings\user>ping 60.0.0.1

Pinging 60.0.0.1 with 32 bytes of data:

Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 60.0.0.1

```

Re-verifying from Win2

Win2 ipconfig — ping 50.0.0.10 unreachable (route not yet added this session)

```

C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 60.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 

C:\Documents and Settings\user>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 50.0.0.10:

```

route add 50.0.0.0 mask 255.0.0.0 60.0.0.1 → ping 50.0.0.10 succeeds

```

C:\WINDOWS\system32\cmd.exe
Default Gateway . . . . . : 
C:\Documents and Settings\user>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>route print

Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 86 5f 09 50 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet
Scheduler Miniport

Active Routes:
=====
Network Destination        Netmask          Gateway           Interface        Metric
60.0.0.0                   255.0.0.0         60.0.0.10         60.0.0.10        10
60.0.0.10                  255.255.255.255   127.0.0.1         127.0.0.1        10
60.255.255.255             255.255.255.255   60.0.0.10         60.0.0.10        10
127.0.0.0                  255.0.0.0         127.0.0.1         127.0.0.1        1
224.0.0.0                  240.0.0.0         60.0.0.10         60.0.0.10        10
255.255.255.255            255.255.255.255   60.0.0.10         60.0.0.10        1

Persistent Routes:
None

C:\Documents and Settings\user>route add 50.0.0.0 mask 255.0.0.0 60.0.0.1

C:\Documents and Settings\user>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time=2ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=2ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 50.0.0.10:

```

Cleanup

Win1 — route delete 60.0.0.0 mask 255.0.0.0 50.0.0.1

```

C:\WINDOWS\system32\cmd.exe
Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\Documents and Settings\user>route delete 60.0.0.0 mask 255.0.0.0 50.0.0.1
C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet
Scheduler Miniport
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
50.0.0.0                   255.0.0.0        50.0.0.10       50.0.0.10        10
50.0.0.10                  255.255.255.255  127.0.0.1       127.0.0.1        10
50.255.255.255             255.255.255.255  50.0.0.10       50.0.0.10        10
127.0.0.0                  255.0.0.0        127.0.0.1       127.0.0.1        1
224.0.0.0                  240.0.0.0        50.0.0.10       50.0.0.10        10
255.255.255.255            255.255.255.255  50.0.0.10       50.0.0.10        1
=====
Persistent Routes:
None
C:\Documents and Settings\user>

```

Result

With no default gateway configured on either PC, ping failed until a manual static route (route add) pointing to the router was added on each host. Once both routes were in place, Win1 and Win2 pinged each other and the router successfully with 0% packet loss.